

Community Detection on Euclidean Random Graphs*

Abishek Sankararaman,[†] François Baccelli[‡]

Abstract

We study Community Detection (CD) on a class of sparse spatial random graphs embedded in the Euclidean space. Our random graph is the planted-partition version of the random connection model studied in Stochastic Geometry. Each node has two labels - an i.i.d. uniform $\{-1, +1\}$ valued community label and a $\mathbb{R}^d$ valued location label which form the support of a Poisson Point Process of intensity λ on $\mathbb{R}^d$. Conditional on the labels, edges are drawn independently at random depending both on the Euclidean distance between the nodes and the community labels on the nodes. The CD problem then consists in estimating the partition of nodes into communities better than at random, based on an observation of the random graph and the spatial location labels on nodes. We establish a non-trivial phase-transition for this problem in terms of λ . We show that for small λ , there exists no algorithm for CD. For large λ , our algorithm solves CD efficiently. We show that for small λ , there exists no algorithm for CD. For large λ , we propose an algorithm which solves CD efficiently. In certain special cases, we establish the exact threshold on λ which separates the existence of an algorithm which solves CD from the impossibility of any algorithm. We also establish a distinguishability result which says that one can always efficiently infer the existence of a partition given the graph and the spatial locations even when one cannot identify the partition better than at random. This is a new phenomenon not observed thus far in any non-spatial Erdős-Rényi based models.

1 Introduction

Community Detection, also known as the graph clustering problem, is the task of grouping together nodes of a graph into representative clusters. This problem has several incarnations that have proven to be useful in various applications ([15]) such as social sciences ([19],[34]), image segmentation [40], recommendation

systems ([22],[37]), web-page sorting [20], and biology ([39], [8]) to name a few. In the present paper, we introduce a new sparse *spatial* random graph and study the graph clustering problem on it. The random graph we consider is denoted by G_n , which has a random number of nodes denoted by N_n that is Poisson distributed with mean λn . In our formulation, n is a scaling parameter going to infinity and $\lambda > 0$ is a *fixed* constant that denotes the *intensity* parameter. Nodes are equipped with two i.i.d. labels, a uniform $\{-1, +1\}$ valued *community label* and a uniform $B_n := \left[-\frac{n^{1/d}}{2}, \frac{n^{1/d}}{2}\right]^d$, $d \in \mathbb{N}$ valued *location label*. Therefore, the average number of nodes having location labels in any subset of B_n of unit volume is λ , which explains why we call it the intensity parameter. Conditional on the node labels, two nodes at locations $x, y \in B_n$ and community labels $Z_x, Z_y \in \{-1, +1\}$ are connected by an edge in G_n independently of other edges with probability $f_{in}(\|x - y\|)$ if $Z_x = Z_y$ or with probability $f_{out}(\|x - y\|)$ if $Z_x \neq Z_y$, where $\|\cdot\|$ denotes the Euclidean norm. We assume $1 \geq f_{in}(r) \geq f_{out}(r) \geq 0$ for all $r \geq 0$ and therefore the graph is expected to have more edges within communities than across. Furthermore, we consider the *sparse* graph regime where $\int_{x \in \mathbb{R}^d} f_{out}(\|x\|) dx \leq \int_{x \in \mathbb{R}^d} f_{in}(\|x\|) dx < \infty$. Hence in this sparse regime, the average degree of any node in G_n is bounded above by $(\lambda/2) \int_{x \in \mathbb{R}^d} (f_{in}(\|x\|) + f_{out}(\|x\|)) dx < \infty$ uniformly in n . Moreover, in this sparse regime, the boundary effects due to the finite window B_n will not matter for large n . We give a rigorous discussion of our model in the sequel in Section 2.

The Community Detection problem in our context refers to when and how one can estimate the partition of the nodes of G_n into communities better than at random, from an observation of the graph **and** the spatial location labels that generated the graph. We say Community Detection is solvable (made precise in Section 2.2) if there exists an algorithm such that the fraction of mis-classified nodes is strictly smaller than a half as n goes to infinity, i.e., we asymptotically beat a random guess of the partition. Although this requirement on

*This work was supported by a grant of the Simons Foundations (#197982) to The University of Texas at Austin

[†]Dept of ECE, The University of Texas at Austin. Email - abishek@utexas.edu

[‡]Department of ECE and Mathematics, The University of Texas at Austin. Email - baccelli@math.utexas.edu.

estimation is very weak, we see through our results that this is indeed the best one can hope for in the sparse graph setting considered here. For simplicity, we also assume that the algorithm has knowledge of the connection functions $f_{in}(\cdot)$, $f_{out}(\cdot)$.

Motivations for a Spatial Model - The most widely studied model for Community Detection is the Stochastic Block Model (SBM), which is a multi-type Erdős-Rényi graph. In the simplest case, the two community symmetric SBM corresponds to a random graph with n nodes, with each node equipped with an i.i.d. uniform community label drawn from $\{-1, +1\}$. Conditionally on the labels, pairs of nodes are connected by an edge independently of other pairs with two different probabilities depending on whether the end points are in the same or different communities. Structurally, the sparse SBM is known to be locally tree-like ([33],[1]) while real social networks are observed to be *transitive* and sparse. Sparsity in social networks can be understood through ‘Dunbar’s number’ [13], which concludes that an average human being can have only about 150 ‘relationships’ (online and offline) at any point of time. Moreover, this is a fundamental cognitive limitation of the person and not that of access or resources, thereby justifying models where average node degree is independent of the population size. Social networks are transitive in the sense that any two agents that share a mutual common neighbor tend to have an edge among them as well, i.e., the graph has many triangles. These observations make the sparse SBM non realistic model and have led to the development of Latent Space Models ([17],[18]) in the social sciences literature. These are sparse *spatial* graphs in which the agents of the social network are assumed to be embedded in an abstract *social space* that is modeled as a finite dimensional Euclidean space and conditional on the embedding, edges of the graph are drawn independently at random as a non-increasing function of the distance between two nodes. Thanks to the properties of Euclidean geometry, these models are transitive and sparse, and have a better fit to data than any SBM ([17]).

Thus, one can view our model as the simplest planted-partition version of the Latent Space model, where the nodes are distributed uniformly in a large compact set B_n and conditional on the locations, edges are drawn depending on Euclidean distance through connection functions $f_{in}(\cdot)$ and $f_{out}(\cdot)$. Although, our assumptions are not particularly tailored towards any real data-sets, our setting is the

most challenging regime for the estimation problem as the location labels alone without the graph reveal no community membership information. However, a drawback in our formulation is that we assume the location labels on nodes are known exactly. In practice, it is likely that the locations labels are unknown (as in the original Latent Space models where the social space is unobservable) or are at-best estimated separately. Nonetheless, our formulation with known location labels forms a crucial first step towards more general models where the location labels are noisy or are missing. Another drawback of our formulation is that we assume the estimator has knowledge of the model parameters $f_{in}(\cdot)$ and $f_{out}(\cdot)$. In our spatial setup, the estimation of connection functions from data is an interesting research question in itself which is however beyond the scope of this paper.

Central Technical Challenges - The core technical challenge in studying our spatial graph model lies in the fact that it is not ‘locally tree-like’. The spatial graph is locally dense (i.e. there are lots of triangles) which arises as a result of the constraints imposed by Euclidean Geometry, while it is globally sparse (i.e. the average degree is bounded above by a constant). The sparse SBM on the other hand, is locally ‘tree-like’ and has very few short cycles [33]. This comes from the fact that the connection probability in a sparse SBM scales as c/n for some $c > 0$. In contrast, the connection function in our model does not scale with n . From an algorithmic point of view however, most commonly used techniques (message passing, broadcast process on trees, convex relaxations, spectral methods etc) are not straight forward to apply to our setting. In contrast, we leverage the observation that our graph is locally dense to propose a simple pairwise clustering scheme (see Algorithm 1) that accurately classifies ‘nearby’ nodes. We then piece together pairwise estimates from nearby nodes for producing the final clustering. To establish a lower bound, we develop a coupling argument by connecting Community Detection with a problem we call ‘Information Flow from Infinity’, which is a new problem and is of independent interest.

The other specialty in our setting is the presence of location labels which are known exactly to the estimator. This provide some form of ‘side-information’ which any estimator must exploit. As an illustrative example to see this, consider the connection functions $f_{in}(\cdot)$ and $f_{out}(\cdot)$ to be of bounded support. In this case, the absence of an edge between two nearby nodes makes it likely that these two nodes belong to

opposite communities but the lack of an edge between far-away nodes farther than the support of either connection functions does not give any community membership information. Thus the lack of an edge in this example has different interpretations depending on the location labels which needs to be exploited in a principled manner by the estimator. Nonetheless, the location labels alone without the graph provide no information on community membership as the community and location labels are independent of each other.

1.1 Overview of our Results and Techniques

The central result in this paper is the existence of a phase-transition for Community Detection on spatial graphs. We show that for every $f_{in}(\cdot), f_{out}(\cdot)$ and $d \geq 2$, there is a non-trivial¹ critical constant $\lambda_c \in (0, \infty)$, such that as λ increases beyond this critical value, the Community Detection problem shifts from being unsolvable to solvable. Our proof for this is in two parts.

1. We establish in Theorem 3.1 that no Community Detection algorithm (polynomial or exponential time) can beat a random guess of the partition if the intensity λ is small (with an explicit estimate provided for this constant). To show this, we first argue that Community Detection is easier than determining whether any two uniformly randomly chosen nodes of G_n belong to the same or different communities with success probability larger than a half. Now since the nodes are uniformly distributed in the set B_n which has volume n , with high probability, the chosen nodes will be ‘far-away’; in particular the distance between the two nodes will exceed any constant r . We then make this pair-wise problem easier by revealing the true label of all nodes at distance r or more from one of the two chosen nodes and asking how well can one estimate the community label on this chosen node. We call this problem ‘Information Flow from Infinity’, since we want to understand whether one can estimate the community label of a uniformly randomly chosen node of G_n , given the true community labels of all nodes ‘far-away’ (at distance r or more). To show impossibility, we construct ‘information percolation’ clusters and establish that when these clusters are ‘small’, then the information from the revealed community labels does not ‘flow’, thereby one cannot solve the pairwise classification problem.

2. On the positive side, we give a novel Community Detection algorithm called GBG in Section 4,

which solves Community Detection if the intensity λ is a sufficiently high constant (with an explicit estimate of the constant). Our algorithm has running time complexity of order n^2 and storage complexity of order n . The key idea is to observe that for any two nodes that are ‘near-by’, we can classify them correctly with exponentially (in λ) small probability of error. However, this is still not sufficient to produce a global clustering since there will be certain pairs incorrectly classified that need to be identified and corrected. We establish this by embedding ‘consistency checks’ into our algorithm to correct some of the mis-classified pairs by partitioning the space B_n into ‘good’ and ‘bad’ regions. We then couple the partitioning of space with another percolation process to prove that our algorithm will mis-classify a fraction strictly smaller than half the nodes if λ is sufficiently high, with an explicit estimate of the constant. Furthermore, in certain special instances of connection functions $f_{in}(\cdot)$ and $f_{out}(\cdot)$, our lower bound and upper bound match to give a sharp phase-transition and we can characterize the critical λ for these cases.

Along the way, in Theorem 3.3 we also consider a problem which is known as the *distinguishability* problem. which asks how well one can solve a hypothesis testing problem between our graph and an appropriate null model (a plain random connection model with connection function $(f_{in}(\cdot) + f_{out}(\cdot))/2$) without communities but having the same average degree and distribution for spatial locations. We show that for *all* parameters, we can solve the distinguishability problem with success probability $1 - o_n(1)$, even if we cannot learn the partition better than a random guess. The key technical contribution lies in identifying suitable graph ‘triangle profiles’ and show that they are different in the planted partition model and the null model. We are able to infer the existence of the partition but do not learn anything about it. This is because we can ‘see’ the partition ‘locally’ in space, but there is no way to consistently piece together the small partitions in different regions of space into one coherent partition of the graph. This phenomenon is new and starkly different from what is observed in the classical Erdős-Rényi based symmetric SBM with two communities where the moment one can identify the presence of a partition, one can also recover the partition better than a random guess ([33]). Moreover, such phenomena where one can infer the existence of a partition but not identify it better than random are conjectured not to occur in the SBM even with many communities [12],[3].

¹The $d = 1$ case is in a sense trivial as seen in Corollary 3.1

1.2 Related Work Community Detection on sparse graphs has mostly been studied on the SBM as the random graph model. The study of SBM has a long history in the different literatures of statistical physics (see [12] and references therein), mathematics (for ex. [5],[33]) and computer science (for ex. [27], [10],[32]). We refer the reader to the comprehensive survey [1] for a complete mathematical and algorithmic treatment of the Community Detection problem on the SBM. The survey of [29] gives a complete treatment of the SBM from a statistical physics view point. There has been renewed interest in the sparse regime of the SBM following the paper of [12], which made number of striking conjectures on phase-transitions. Subsequently, some of them have been established with the most notable and relevant achievements to ours being that of [33],[31], [26] and [7]. These papers prove that both Community Detection and the distinguishability problem for the two community sparse SBM undergo a phase-transition at the same point which they characterize explicitly. These results for the SBM motivates the investigation of the phase-transitions for Community Detection and distinguishability in our model. However, the tools needed for our model are very different from those used to study the SBM. Most prior work on sparse SBM relies on the locally tree-like property of the random graph in some way or the other ([1] has a nice account of the methods) and are hence not directly applicable in our setting. Thus, the key ideas for all of our results come from different problems, mostly those studied in the theory of percolation [6] and stochastic geometry [35]. Some of the ideas for the analysis of our algorithm bears some connections to ideas that appeared in the Interacting Particle Systems literature (for ex. [35] [21],[36],[14]). Precisely, the dependent percolation process we use in our analysis was also applied in studying other problems originating in statistical physics. Nevertheless, the application of this in the context of algorithm design is new to the best of our knowledge. The lower bound comes from identifying an *easier* problem than Community Detection called Information Flow from Infinity, which is a new problem. The key idea to study this comes from certain ‘random-cluster method’ and coupling arguments. Such methods are quite popular and have been developed extensively in other contexts for example to study mixing time of Ising Models ([25],[24],[23]). Coupling ideas similar in spirit to ours has also appeared in other estimation contexts such as the reconstruction on trees problem [30] where a lower bound was established using this

method. However, in all of the previous literature, this coupling method was applied on fixed graphs (grids, trees, bounded degree graphs), while we apply it to the continuum (the spatial locations are not discrete). From a modeling perspective, the work of [42] which considers the Latent Space Models as a part of its model is the closest to our model. They give a spectral algorithm that is proven to work in the logarithmic degree regime, which is different from the sparse regime considered in the present paper.

In parallel independent work, a similar model to ours was also recently investigated by [16], where they again consider the logarithmic regime. They study a setting in which the location labels on nodes are not known, but only the graph is known. Their algorithm is quite similar to the one presented here, wherein they also exploit the fact that the spatial graph is ‘locally-dense’. A challenging future work now will be to consider the sparse regime in their setting where the location labels are also missing.

1.3 Organization of the paper We give a formal description of the model and the problem statement in Section 2. We then present our main theorem statements in Section 3. The subsequent sections will develop the ideas and the proofs needed for our main results. We describe our GBG Algorithm in Section 4 where we first give the idea and then the details of the algorithm. The analysis of our algorithm is performed in Section 5, where the key idea is to construct coupling arguments with site percolations. We establish the lower bound in Section 6, where we first introduce the Information Flow from Infinity problem and then prove that this is easier than Community Detection. Subsequently, we provide a proof of the impossibility result for Information Flow from Infinity. In Section 7, we consider the distinguishability problem and provide a proof of it.

2 Mathematical Framework and Problem Statement

We describe the mathematical framework based on stationary point processes and state the problem of Community Detection. We just set a common shorthand notation we use throughout the paper. For an arbitrary $t \in \mathbb{R}_+$, we denote by the term $o_t(1)$ to be a bounded function $t \rightarrow a_t$ from $\mathbb{R}_+$ to $\mathbb{R}_+$ such that $\lim_{t \rightarrow \infty} a_t = 0$.

2.1 The Planted Partition Random Connection Model For technical simplicity, we first define an infinite spatial random graph G and consider

the Community Detection problem on appropriate finite sub-graphs G_n . We suppose there exists an abstract probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which we will construct our *infinite* random graph G which is parametrized by $\lambda \in \mathbb{R}_+$, $d \geq 2$ and two functions $f_{in}(\cdot), f_{out}(\cdot) : \mathbb{R}_+ \rightarrow [0, 1]$ such that for all $r \geq 0$ we have $f_{in}(r) \geq f_{out}(r)$. The nodes of G are indexed using $\mathbb{N}$ and each node has two labels, one denoting its location in $\mathbb{R}^d$, and another being a community label taking values in $\{-1, +1\}$. We first sample the location and community labels of G by a marked point-process, where atoms of the point process represent the location labels on nodes and marks of the point-process represent community labels on nodes. Conditionally on the locations and the community labels, the edges of G will be drawn independently at random between pairs of nodes. Formally, we let $\phi := \{X_1, X_2, \dots\}$ with $X_i \in \mathbb{R}^d$ for all $i \in \mathbb{N}$, to be a Poisson Point Process (PPP)² of intensity λ on $\mathbb{R}^d$. We further assume that $\phi = \{X_1, X_2, \dots\}$ is enumerated in increasing l_∞ distance, i.e. for all $i \in \mathbb{N}$, $\|X_{i+1}\|_\infty \geq \|X_i\|_\infty$. The interpretation is that node $i \in \mathbb{N}$ of G has for its location label, the random variable $X_i \in \mathbb{R}^d$. We further mark each atom $i \in \mathbb{N}$ of ϕ with random variables $Z_i \in \{-1, +1\}$ and $\{U_{ij}\}_{j \in \mathbb{N} \setminus \{i\}} \in [0, 1]^{\mathbb{N} \setminus \{i\}}$ satisfying $U_{ij} = U_{ji}$ for all $i \neq j \in \mathbb{N}$. The sequence $\{Z_i\}_{i \in \mathbb{N}}$ is i.i.d. with each element being uniformly distributed in $\{-1, +1\}$. For every $i \in \mathbb{N}$, the sequence $\{U_{ij}\}_{j \in \mathbb{N} \setminus \{i\}}$ is i.i.d. with each element of $\{U_{ij}\}_{j \in \mathbb{N} \setminus \{i\}}$ being uniformly distributed on $[0, 1]$. The interpretation of ϕ and its marks is that every node $i \in \mathbb{N}$ of G has location label given by $X_i \in \mathbb{R}^d$ and a community label given by $Z_i \in \{-1, +1\}$. This community label is moreover i.i.d. and independent of everything else. We will use the marks $\{U_{ij}\}_{j \in \mathbb{N} \setminus \{i\}}$ to sample the graph neighbors of a node i as follows. An edge between two nodes i and j where $i \neq j$ is present in G if and only if $U_{ij} = U_{ji} \leq f_{in}(\|X_i - X_j\|)\mathbf{1}_{Z_i=Z_j} + f_{out}(\|X_i - X_j\|)\mathbf{1}_{Z_i \neq Z_j}$. In other words, conditionally on the node labels, an edge is present between node i and j independent of other pair of edges with probability $f_{in}(\|X_i - X_j\|)$ if $Z_i = Z_j$ or with probability $f_{out}(\|X_i - X_j\|)$ if $Z_i \neq Z_j$. From Campbell's theorem, the average number of neighbors any node has from its own community is $(\lambda/2) \int_{x \in \mathbb{R}^d} f_{in}(\|x\|) dx$ and $(\lambda/2) \int_{x \in \mathbb{R}^d} f_{out}(\|x\|) dx$ in the opposite community. Hence, if $\int_{x \in \mathbb{R}^d} f_{in}(\|x\|) dx < \infty$, almost-surely, all nodes have finite degree and we call this regime as

sparse. In the rest of this paper, we will assume that $\int_{x \in \mathbb{R}^d} f_{in}(\|x\|) dx < \infty$, i.e. the graph is sparse unless we state otherwise.

The random graph G can be viewed as a ‘planted-partition’ version of the classical random-connection model ([28]). Given $\lambda \in \mathbb{R}_+$, $g(\cdot) : \mathbb{R}_+ \rightarrow [0, 1]$ and $d \geq 1$, the classical random-connection model $H_{\lambda, g(\cdot), d}$, is a random graph whose vertex set forms a homogeneous PPP of intensity λ on $\mathbb{R}^d$. Conditionally on the locations, edges in $H_{\lambda, g(\cdot), d}$ are placed independently of each other where two points at locations x and y of $\mathbb{R}^d$ are connected by an edge in $H_{\lambda, g(\cdot), d}$ with probability $g(\|x - y\|)$. This construction can be made precise by letting the edge random variables for each node be marks of the PPP similarly to the construction of G .

2.2 The Community Detection Problem To state the Community Detection problem, we will consider appropriate finite restrictions of the infinite random graph G . Denote by $B_n := \left[-\frac{n^{1/d}}{2}, \frac{n^{1/d}}{2}\right]^d$, the cube of area n in $\mathbb{R}^d$. Let ϕ_n be the restriction of ϕ to B_n and G_n be the restriction of G to the nodes that are located in B_n . Denote by N_n the number of nodes of the graph G_n . From properties of the PPP, the random variable N_n is Poisson distributed with mean λn . Moreover, conditionally on N_n , these nodes are placed uniformly and independently in B_n . Recall that since the atoms of ϕ were numbered in increasing l_∞ distance, $N_n = \inf\{i \in \mathbb{N} : X_i \notin B_n\}$, i.e. the first N_n atoms of ϕ lie in B_n . Notice that the description given here and in the introduction are identical in distribution and hence we decide to use the same notation as the problem definition is only distributional as seen in Definition 2.2 below. Before describing the problem, we need the definition of ‘overlap’ between two sequences.

DEFINITION 2.1. Given a $t \in \mathbb{N}$, and two sequences $\mathbf{a}, \mathbf{b} \in \{-1, 1\}^t$, the overlap between $\mathbf{a}$ and $\mathbf{b}$ is defined as $\frac{|\sum_{i=1}^t a_i b_i|}{t}$, i.e. the absolute value of the normalized scalar product.

DEFINITION 2.2. Community Detection is said to be **solvable** for $\lambda, d, f_{in}(\cdot)$ and $f_{out}(\cdot)$ if for every $n \in \mathbb{R}_+$, there exists a sequence of $\{-1, +1\}$ valued random variables $\{\tau_i^{(n)}\}_{i=1}^{N_n}$ which is a deterministic function of the observed data G_n **and** ϕ_n such that there exists a constant $\gamma > 0$ satisfying

$$(2.1) \quad \liminf_{n \rightarrow \infty} \mathbb{P}[\mathcal{O}_n \geq \gamma] = 1,$$

²This is defined in Appendix ??

where $\mathcal{O}_n$ is the overlap between $\{\tau_i^{(n)}\}_{i=1}^{N_n}$ and $\{Z_i\}_{i=1}^{N_n}$.

Note that we assume that the algorithm has access to the parameters $f_{in}(\cdot)$, $f_{out}(\cdot)$ and λ although this assumption may not always hold in practice. As mentioned, the estimation of model parameters from data itself will form an interesting technical question which we leave for future work. We take an absolute value in the definition of overlap since the distribution of G is symmetric in the community labels. In particular, if we flipped all community labels of G , we would observe a graph which is equal in distribution to G . Thus, any algorithm can produce the clustering only up-to a global sign flip, which we capture by considering the absolute value. We take finite restrictions B_n since the overlap is not well defined if $N_n = \infty$. A natural question then is of ‘boundary-effects’, i.e. the nodes near the boundary of B_n will have different statistics for neighbors than those far away from the boundary. However since G_n is sparse, except for a $o_n(1)$ fraction of nodes, all nodes in G_n will have the same degree as in the infinite graph G , i.e. the boundary effects are negligible.

A key new feature of our Definition 2.2 comes from our assumption that the algorithm has knowledge of all location labels on the nodes and it only needs to estimate the missing community labels. Our definition of detection which sometimes is also referred to as weak-recovery in the literature, deems an algorithm successful at community detection if it can correctly classify a fraction larger than half of the nodes. Observe that achieving $\gamma = 0$ (which amounts to correctly classifying exactly half the nodes) in the definition is trivial: this is obtained by associating a +1 to all nodes; by the Strong-Law of Large numbers, we will achieve an overlap of 0. There are other notions of recovery of interest such as strong recovery or exact-recovery which asks whether one can achieve $\gamma = 1 - \epsilon$ for all $\epsilon > 0$ or $\gamma = 1$ respectively. However, if the graph G is sparse, we show (in Corollary 3.1), that achieving $\gamma = 1 - \epsilon$ for all $\epsilon > 0$ is not possible. Thus, the interesting question in the sparse regime is to consider when and how one can achieve any $\gamma > 0$.

The following elementary monotonicity property is evident from the definition of the problem and sets the stage for stating our main results.

PROPOSITION 2.1. *For every $f_{in}(\cdot), f_{out}(\cdot)$ and d , there exists a $\lambda_c \in [0, \infty]$ such that*

- $\lambda < \lambda_c \Rightarrow$ Community Detection is not solvable.
- $\lambda > \lambda_c \Rightarrow$ there exists an algorithm (which could possibly take exponential time) to solve Community Detection.

Proof. Assume, for a given value of λ , there exists a community detection algorithm that achieves an overlap of $\gamma > 0$. Now, given any $\lambda' > \lambda$, we will argue that we can achieve positive overlap. The proof of this follows from the basic thinning properties of the PPP. Given an instance of the problem with intensity λ' , we will remove every node along with its incident edges independently with probability $1 - \frac{\lambda}{\lambda'}$. We assign a community label estimate of +1 to all the removed nodes. For the nodes that remain (which is then an instance of the problem of community detection with intensity λ), we achieve an overlap of γ from the hypothesis that we can achieve positive overlap at intensity λ . Thus, from the independence of the thinning procedure and the community labels and strong law of large numbers, the overlap achieved by this process on an instance of intensity λ' will be at-least $\frac{\lambda\gamma}{\lambda'}$ which is strictly positive. Thus, the problem of community detection solvability is monotone in λ .

This proposition is not that strong since it does not rule out the fact that λ_c is either 0 or infinity. Moreover, this proposition does not tell us anything about polynomial time algorithms, just of the existence or non-existence of any (polynomial or exponential time) algorithms. The first non-trivial result would be to establish that $0 < \lambda_c < \infty$, i.e. the phase transition is strictly non-trivial and then to show that for possibly a larger constant, the problem is solvable efficiently. We establish both of this in Section 3.

Notation - Palm Probability Before stating the results, we will need an important definition. We define the Palm Probability measure $\mathbb{P}^0$ of a point process in this subsection for future reference. Roughly stated, the Palm measure is the distribution of a marked point process ϕ ‘seen from a typical atom’. We refer the reader to [11] for the general theory of Palm measures. However thanks to Slivnyak’s theorem [11], we have a simple interpretation of $\mathbb{P}^0$ which is what we will use. The measure $\mathbb{P}^0$ is obtained by first sampling ϕ and G from $\mathbb{P}$ and placing an additional node indexed 0 at the origin of $\mathbb{R}^d$ and equipping it with independent community label and edges. The label of this node at origin will be denoted by $Z_0 \in \{-1, +1\}$ which is uniform and independent of

anything else. Conditionally on Z_0 , ϕ and the labels $\{Z_i\}_{i \in \mathbb{N}}$, we place an edge between node $i \in \mathbb{N}$ and this extra node at the origin with probability $f_{in}(\|X_i\|)\mathbf{1}_{Z_i=Z_0} + f_{out}(\|X_i\|)\mathbf{1}_{Z_i=-Z_0}$ independently of the edges between $j \neq i \in \mathbb{N}$ and the origin.

3 Main Results

3.1 Lower Bound for Community Detection

To state the main lower bound result in Theorem 3.1 below, we set some notation. For the random connection model graph $H_{\lambda,g(\cdot),d}$, denote by $\mathcal{C}_{H_{\lambda,g(\cdot),d}}(0)$ the set of nodes of $H_{\lambda,g(\cdot),d}$ that are in the same connected component as that of the node at the origin under the measure $\mathbb{P}^0$. Denote by $\theta(H_{\lambda,g(\cdot),d}) := \mathbb{P}^0[|\mathcal{C}_{H_{\lambda,g(\cdot),d}}(0)| = \infty]$ the percolation probability of the random graph $H_{\lambda,g(\cdot),d}$, i.e. the probability (under Palm) that the connected component of the origin has infinite cardinality.

THEOREM 3.1. *If $\theta(H_{\lambda,f_{in}(\cdot)-f_{out}(\cdot),d}) = 0$, then Community Detection is not solvable.*

This theorem states that if the two functions $f_{in}(\cdot)$ and $f_{out}(\cdot)$ are not ‘sufficiently far-apart’, then no algorithm to detect the partition of nodes can beat a random guess. As a corollary, this says that Community Detection is impossible for $d = 1$.

COROLLARY 3.1. *For all $\lambda > 0$, $f_{in}(\cdot), f_{out}(\cdot)$ such that $\int_{x \in \mathbb{R}} f_{in}(\|x\|)dx < \infty$, Community Detection is not solvable if $d = 1$.*

Proof. This is based on the classical fact that for all $g(\cdot) : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $\int_{x \in \mathbb{R}} g(\|x\|)dx < \infty$, $\theta(H_{\lambda,g(\cdot),1}) = 0$.

The following corollary gives a quantitative estimate of the percolation probability for higher dimensions in terms of the problem parameters.

COROLLARY 3.2. *For all $d \geq 2$, if $\lambda \leq \lambda_{lower} := (\int_{x \in \mathbb{R}^d} (f_{in}(\|x\|) - f_{out}(\|x\|))dx)^{-1}$, then Community Detection cannot be solved. Thus, $\lambda_c > (\int_{x \in \mathbb{R}^d} (f_{in}(\|x\|) - f_{out}(\|x\|))dx)^{-1}$.*

Proof. From classical results on percolation [28], by comparison with a branching process, we see that $\lambda \int_{x \in \mathbb{R}^d} g(\|x\|) \leq 1 \Rightarrow \theta(H_{\lambda,g(\cdot),d}) = 0$.

Recall that if the graph G is sparse (finite average degree), then $\int_{x \in \mathbb{R}^d} f_{in}(\|x\|)dx < \infty$ which implies from Corollary 3.2 that λ_c is strictly positive in the sparse regime. The following proposition shows that this lower bound is tight for certain specific families of connection functions.

PROPOSITION 3.1. *For all $d \geq 2$ and $R_1 \geq R_2$, if $f_{in}(r) = \mathbf{1}_{r \leq R_1}$ and $f_{out}(r) = \mathbf{1}_{r \leq R_2}$ and λ is such that $\theta(H_{\lambda,f_{in}(\cdot)-f_{out}(\cdot),d}) > 0$, then community detection can be solved in time proportional to n with the proportionality constant depending on the parameters λ, R_{in} and R_{out} .*

Hence, in view of Theorem 3.1, for $f_{in}(r) = \mathbf{1}_{r \leq R_1}$ and $f_{out}(r) = \mathbf{1}_{r \leq R_2}$ for some $R_1 \geq R_2$, then if λ is such that $\theta(H_{\lambda,f_{in}(\cdot)-f_{out}(\cdot),d}) = 0$, no algorithm (exponential or polynomial) time can solve Community Detection, while if λ is such that $\theta(H_{\lambda,f_{in}(\cdot)-f_{out}(\cdot),d}) > 0$, then a linear time algorithm exists to solve Community Detection. This gives a sharp phase-transition for this particular set of parameters where the problem shifts from being unsolvable even with unbounded computation to being efficiently solvable.

Algorithm and an Upper Bound for Community Detection Our main result in the positive direction is our GBG algorithm described in Section 4. The main theorem statement on the performance of GBG is the following.

THEOREM 3.2. *If $f_{in}(\cdot)$ and $f_{out}(\cdot)$ are such that $\{r \in \mathbb{R}_+ : f_{in}(r) \neq f_{out}(r)\}$ has positive Lebesgue measure and $d \geq 2$, then there exists a $\lambda_{upper} < \infty$ depending on $f_{in}(\cdot), f_{out}(\cdot)$ and d , such that for all $\lambda \geq \lambda_{upper}$, the GBG algorithm solves the Community Detection problem. Moreover, GBG when run on data (G_n, ϕ_n) , has time complexity order n^2 and storage complexity order n .*

REMARK 3.1. *An upper bound to λ_{upper} is given in Proposition 5.6.*

This gives a complete non-trivial phase-transition for the sparse graph case where we have $0 < \lambda_c < \infty$ which implies the existence of different phases. Moreover, we have quantitative bounds $\lambda_{lower} \leq \lambda_c \leq \lambda_{upper}$ for the critical value where the phase-transition occurs.

3.2 Identifiability of the Planted Partition

The key result we show here is that unlike in the traditional Erdős-Rényi setting, the planted partition random connection model is always mutually singular with respect to *any* random connection model without communities. Before precisely stating the result, we set some notation. Denote by $\mathbb{M}_G(\mathbb{R}^d)$ the Polish space of all simple spatial graphs whose vertex set forms a locally finite set of $\mathbb{R}^d$. Thus, our random graph G or the random connection model $H_{\lambda,g(\cdot),d}$

can also be viewed through the induced measure on the space $\mathbb{M}_G(\mathbb{R}^d)$.

THEOREM 3.3. *For every $\lambda > 0$, d and connection functions $f_{in}(\cdot)$ and $f_{out}(\cdot)$ satisfying $1 \geq f_{in}(r) \geq f_{out}(r) \geq 0$ for all $r \geq 0$, and $\{r \geq 0 : f_{in}(r) \neq f_{out}(r)\}$ having positive Lebesgue measure and $g(\cdot) : \mathbb{R}_+ \rightarrow [0, 1]$, the probability measures induced on the space of spatial graphs $\mathbb{M}_G(\mathbb{R}^d)$ by G and $H_{\lambda, g(\cdot), d}$ are mutually singular.*

This theorem gives in particular that G and $H_{\lambda, \frac{f_{in}(\cdot) + f_{out}(\cdot)}{2}, d}$ are mutually singular. Note that if $g(\cdot) \neq \frac{f_{in}(\cdot) + f_{out}(\cdot)}{2}$, then the average degrees of G and $H_{\lambda, g(\cdot), d}$ are different and hence the empirical average of the degrees in G_n and $H_{\lambda, g(\cdot), d}$ restricted to B_n , will converge almost surely as $n \rightarrow \infty$ (thanks to the ergodic property of PPP) to the mean degree, thereby making the two induced measures mutually singular. Thus, the only non-trivial random connection model that can possibly be not singular with respect to G is $H_{\lambda, \frac{f_{in}(\cdot) + f_{out}(\cdot)}{2}, d}$, i.e. the case of equal average degrees. We show by a slightly different albeit similar ergodic argument in Section 7 that even in the case of equal average degrees, the two induced measures are mutually singular.

This theorem bears on the *distinguishability* problem which is a hypothesis testing question where one needs to predict whether the data (graph along with spatial locations) is drawn from G or $H_{\lambda, \frac{f_{in}(\cdot) + f_{out}(\cdot)}{2}, d}$ with a success probability exceeding a half when the prior distribution is uniform over the two models. Theorem 3.3 implies that this hypothesis testing problem can be solved with probability of success $1 - o_n(1)$ for *all* parameter values. Thus, the distinguishability problem as stated in our spatial case exhibits no phase-transition. A consequence of our results is that in certain regimes ($\lambda < \lambda_c$ for $d \geq 2$ and $\lambda > 0$ for $d = 1$), we can be very sure by observing the data that a partition exists, but cannot identify it better than at random. Such phenomena was proven not to be observed in a symmetric SBM with two communities ([33]) and conjectured not to occur in any arbitrary SBM ([12]).

4 Algorithm for Performing Community Detection

In this section, we outline an algorithm called GBG described in Algorithm 3 that has time complexity of order n^2 and storage complexity of order n and solves the Community Detection problem for all sufficiently

high λ .

4.1 Key Idea behind the Algorithm - Dense Local Interactions

The main and simple idea in our algorithm is that the graph G_n is ‘locally-dense’ even though it is globally sparse. This is in contrast to sparse Erdős-Rényi based graphs in which the local neighborhood of a typical vertex ‘looks like a tree’, our graph will have a lot of triangles due to Euclidean geometry. This simple observation that our graph is locally-dense enables us to propose simple pairwise estimators as described in Algorithm 1 which exploits the fact that two nodes ‘nearby’ in space have a lot of common neighbors (order λ). For concreteness, consider the case when $f_{in}(r) = a\mathbf{1}_{r \leq R}$ and $f_{out}(r) = b\mathbf{1}_{r \leq R}$ for some $R > 0$ and $0 \leq b < a \leq 1$. This means that points at Euclidean distance of R or lesser are connected by an edge in G with probability either a or b depending on whether the two points have the same community label or not. Moreover from elementary calculations, the number of common graph neighbors for any two nodes of G at a distance αR away for some $\alpha < 2$ is a Poisson random variable with mean either $\lambda c(\alpha)R^d(a^2 + b^2)/2$ or $\lambda c(\alpha)R^d ab$ (for some constant $c(\alpha)$ that comes from geometric arguments) depending on whether the two nodes have the same or different community labels. Thus, using a simple strategy consisting of counting the number of common neighbors and thresholding gives a probability of mis-classifying any ‘nearby’ pair of nodes to be exponentially small in λ . We implement this simple idea in the sub-routine 1 below. Now, to produce the global partition one needs care to aggregate the pairwise estimates into a global partition. Since some pair-wise estimates are bound to be in error, we must identify them and avoid using those erroneous pair-wise estimates (see also Figure 1). We achieve this by classifying regions of space B_n as ‘good’ or ‘bad’ and then by considering the pair-wise estimates only in the ‘good’ regions. We prove that if λ is sufficiently large, then the ‘good’ regions will have sufficiently large volume and hence will succeed in detecting the communities better than at random.

4.2 Notations and Definitions In this section, we specify the needed notations for describing our algorithm. We will assume that the connection functions $f_{in}(\cdot)$ and $f_{out}(\cdot)$ satisfy the hypothesis of Theorem 3.2. Thus, there exists $0 \leq \tilde{r} < R < \infty$ such that $f_{in}(r) > f_{out}(r)$ for all $r \in [\tilde{r}, R]$. In the

rest of this section, we will use the $\tilde{r}$ and R coming from the connection functions.

To describe the algorithm, we need to set some notation. We partition the entire infinite domain $\mathbb{R}^d$ into good and bad regions. However this is just for simplicity and in practice, it suffices to do the partition for the region B_n . We first tessellate the space $\mathbb{R}^d$ into cubes of side-length $\frac{R}{4d^{1/d}}$ where R is as above. We identify the tessellation with the index set $\mathbb{Z}^d$, i.e. the cell indexed z is a cube of side-length $\frac{R}{4d^{1/d}}$ centered at the point $\frac{zR}{4d^{1/d}} \in \mathbb{R}^d$. The subset of $\mathbb{R}^d$ that corresponds to cell z is denoted by Q_z . Hence the cell indexed 0 is the cube of side-length $\frac{R}{4d^{1/d}}$ centered at the origin. We now give several definitions on the terminology used for the $\mathbb{Z}^d$ tessellation and not to be confused with the terminology for describing the graph G_n . We collect all the different notation and terminology in this sub-section for easier access and reference.

DEFINITION 4.1. A set $U \subseteq \mathbb{Z}^d$ is said to be $\mathbb{Z}^d$ -**connected** if for every $x, y \in U$, there exists a $k \in \mathbb{N}$ and $x_1, \dots, x_k \in U$ such that for all $i \in [0, k+1]$, $\|x_i - x_{i-1}\|_\infty = 1$, where $x_0 := x$ and $x_{k+1} := y$.

DEFINITION 4.2. For any $z \in \mathbb{Z}^d$, denote by $\mathbb{Z}^d$ -**neighbors** of z the set of all $z' \in \mathbb{Z}^d$ such that $\|z - z'\|_\infty \leq 1$.

DEFINITION 4.3. For any subset $A \subset \mathbb{Z}^d$ and any $k \in \mathbb{N}$, the k **thickening** of A is denoted by $\mathbf{L}_k(A) := \bigcup_{z \in A} \bigcup_{z' \in \mathbb{Z}^d: \|z - z'\|_\infty \leq k} z'$.

DEFINITION 4.4. For any set $B \subseteq \mathbb{Z}^d$, denote by the set $Q_B := \bigcup_{z \in B} Q_z$.

DEFINITION 4.5. Let $\mathcal{Z}(\cdot) : \mathbb{R}^d \rightarrow \mathbb{Z}^d$ be the projection function, i.e. $\mathcal{Z}(x) := \inf\{z \in \mathbb{Z}^d : \|\frac{Rx}{4d^{1/d}} - z\|_\infty \leq 0.5\}$. In case, of more than one z achieving the minimum, we take the lexicographically smallest such z .

DEFINITION 4.6. For any two points $x, y \in \mathbb{R}^d$, denote by $S_R(x, y) := B(x, R) \cap B(y, R)$, i.e. the intersection of two balls of radius R centered at points x and y .

DEFINITION 4.7. For any two points $x, y \in \mathbb{R}^d$ such that $\|x - y\|_2 < R$, define by the two constants $M_{in}(x, y)$ and $M_{out}(x, y)$ as follows.

$$M_{in}(x, y) = \int_{z \in S_R(x, y)} \left(f_{in}(\|x - z\|) f_{in}(\|y - z\|) + f_{out}(\|x - z\|) f_{out}(\|y - z\|) \right) dz$$

$$M_{out}(x, y) = \int_{z \in S_R(x, y)} \left(f_{in}(\|x - z\|) f_{out}(\|y - z\|) + f_{out}(\|x - z\|) f_{in}(\|y - z\|) \right) dz$$

Observe that the definitions of $M_{in}(x, y)$ and $M_{out}(x, y)$ immediately give that

$$M_{in}(x, y) - M_{out}(x, y) = \int_{z \in S_R(x, y)} \left(f_{in}(\|x - z\|) - f_{out}(\|x - z\|) \right) \left(f_{in}(\|y - z\|) - f_{out}(\|y - z\|) \right) dz.$$

DEFINITION 4.8. For any two points $x, y \in \phi$, denote by $E_G^{(R)}(x, y)$ the number of common graph neighbors of x and y in G which are within a distance R from both x and y .

4.3 Algorithm Description in Pseudo Code

We first present two sub-routines in Algorithms 1 and 2 that classify each cell of $\mathbb{R}^d$ to be either Good or Bad. The algorithm is parametrized by $\epsilon \in (0, \frac{1}{2})$ which is arbitrary and fixed.

Algorithm 1 Pairwise Classifier

```

1: procedure PAIRWISE-CLASSIFY( $i, j, \phi, G$ )
2:   if  $E_G^{(R)}(X_i, X_j) > \frac{\lambda}{2} (M_{in}(X_i, X_j) + M_{out}(X_i, X_j))$  then return 1
3:   else
4:     return -1
5:   end if
6: end procedure

```

In this algorithm, we classify two nodes as in the same partition if the number of common graph neighbors they have exceeds a threshold. The threshold is the average of the expected number of neighbors if the two nodes in consideration are of the same or opposite communities. Such simple tests suffices for our purpose, although one could imagine a more accurate estimator that also takes into account the number of nodes in $S_R(X_i, X_j)$ that do not have any edges to X_i and X_j ; or the location labels of the common neighbors.

Algorithm 2 Is A-Good Testing

```

1: procedure IS-A-GOOD( $z, G$ )
2:   if  $|\phi \cap Q_z| < \lambda(R/4)^d(1/d)(1-\epsilon)$  then return
     FALSE
3:   end if
4:    $\phi^{(z)} := \phi \cap (\cup_{z': \|z-z'\|_\infty \leq 1} Q_{z'})$ 
5:   for all  $\forall k \geq 1$ , and all  $X_1, \dots, X_k \in \phi^{(z)}$  do
6:     if  $\prod_{i=1}^k \text{PAIRWISE-CLASSIFY}(X_i, X_{i+1}, G) = -1$  then  $\triangleright$  Where
        $X_{k+1} := X_1$ 
7:       return FALSE
8:     end if
9:   end for
10:  return TRUE
11: end procedure

```

Algorithm 3 GBG

```

1: procedure MAIN-ROUTINE( $G_n, \phi_n$ )
2:   Classify each cell in  $B_n$  to be either A-Good
   or A-Bad using subroutine Is-A-Good.
3:   Let  $\mathcal{D}_1, \dots, \mathcal{D}_k$  be the A-Good  $\mathbb{Z}^d$ -connected
   components in  $B_n$ .
4:   for  $l = 1, l \leq k$  do
5:     Let  $X_{l_1}, \dots, X_{l_{n_l}} \in \phi_n \cap Q_{\mathcal{D}_j}$  be maxi-
     mal and arbitrary s.t  $\|\mathcal{Z}(X_{l_o}) - \mathcal{Z}(X_{l_{o+1}})\|_\infty \leq$ 
      $1, \forall 1 \leq o \leq n_l - 1$ 
6:     Set  $\hat{\tau}_{l_1}^{(n)} = +1$ 
7:     for  $w = 2, w \leq n_l$  do
8:       Set  $\hat{\tau}_{l_w}^{(n)} = \text{Pairwise-}$ 
       Classify( $l_{w-1}, l_w, \phi_n, G_n$ ) $\hat{\tau}_{l_w}^{(n)}$ 
9:     end for
10:   end for
11:   for  $c = 1, c \leq N_n$  do
12:     if  $\hat{\tau}_c^{(n)} = 0$  then
13:       Set  $\hat{\tau}_c^{(n)} = +1$ 
14:     end if
15:   end for
16:   return  $\{\hat{\tau}_i^{(n)}\}_{i=1}^{N_n}$ 
17: end procedure

```

To understand the algorithm, we need some definitions which classify cells of $\mathbb{Z}^d$ into Good or Bad depending on the ‘local graph geometry’.

DEFINITION 4.9. A cell Q_z is **A-Good** if

1. $|\phi \cap Q_z| \geq \lambda \left(\frac{R}{4}\right)^d \frac{1}{d}(1-\epsilon)$; and
2. **Is-A-Good**(z, G) returns **TRUE**

A cell is called **A-Bad** if it is not A-Good.

The key idea of our simple algorithm lies in the definition of A-Good cells. We classify a cell to be A-Good if there are no ‘inconsistencies’ in the Pairwise-Estimates. See Figure 1 for an example of pair-wise inconsistency due to the **Pairwise-Classify** algorithm. In words, a cell is A-Good, if among the nodes of G that either lie in the cell under consideration or in the neighboring cells, there are no inconsistencies in the output returned by the **Pairwise-Classify** algorithm. Moreover, one can test whether a cell is A-Good or not based on the data (ϕ, G) itself as done in Algorithm 2. Thus, we use the nomenclature of *Algorithm-Good* as A-Good.

The main routine in Algorithm 3 proceeds as follows. In Line 3, we extract out all A-Good connected cells in the spatial region B_n . Suppose that there are k A-Good connected components denoted by $\mathcal{D}_1, \dots, \mathcal{D}_k$. Our algorithm looks at each connected component independently and produces a labeling of the nodes in them. In Line 5, we enumerate all nodes in any A-Good connected component $\mathcal{D}_l$ as $X_{l_1} \dots X_{l_{n_l}}$ such that for all $1 \leq o < n_l$, we have $\|\mathcal{Z}(X_{l_o}) - \mathcal{Z}(X_{l_{o+1}})\|_\infty \leq 1$. Such an enumeration of any A-Good connected component is possible since by definition, every A-Good cell is non-empty of nodes. Now, we sequentially estimate the community labels in Line 8 using the **Pairwise-Classify** sub-routine applied on ‘nearby’ pairs of nodes. In Line 13, we assign an estimate of +1, i.e. extract no meaningful clustering for nodes that fall in A-Bad cells. See also Figure 2 for an illustration.

4.4 Complexity and Implementation We discuss a simple implementation of our algorithm which takes time of order n^2 to run and storage space of order n . The multiplicative constants here depend on λ . We store the locations ϕ_n as a vector whose length is order λn and the graph G_n as an adjacency list. An adjacency list representation is appropriate since G_n is sparse and the average degree of any node is a constant (that depends on λ). Once we sample the locations ϕ_n , the graph G_n takes time of order n^2 to sample. However, if one represented the locations of nodes more cleverly, then the sampling complexity could possibly be reduced from n^2 . Moreover, since the average degree is a constant, the storage needed is order n . Given the data ϕ_n and the graph G_n , we pre-process this to store another adjacency list where for every vertex, we store the list of all other

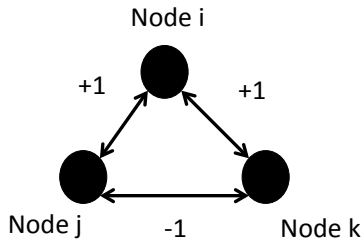


Figure 1: An illustration where Pairwise-Classify leads to inconsistency. The values on the edges represent the output of pairwise classify run on the two end points as inputs. In this example it is clear that for at-least one pair (i, j) , (j, k) , (k, i) , the output of pairwise estimate is different from the ground truth.

vertices within a distance of $2R$ from it. This preprocessing takes order n^2 time and order n space. The space complexity is order n since the graph is sparse. Equipped with this, we create a ‘grid-list’ where for each coordinate of $\mathbb{Z}^d$, we store the list of vertices whose location is in the considered grid cell. This takes just order n time to build. Moreover, since only a constant number of nodes are in any grid cell and the set B_n contains order n cells, the storage space needed for ‘grid-list’ is order n . Furthermore, since only a constant number of nodes are in a cell, it takes a constant time to test whether a particular cell is A-Good or A-Bad. Thus, to find $\mathbb{Z}^d$ connected components of Good-cells and produce the clustering takes another order nd time where d is the dimension. This gives our algorithm overall a time complexity of order n^2 and a storage complexity of order n .

5 Analysis and Proof of the Algorithm

The following theorem is the main theoretical guarantee on the performance of the GBG algorithm.

THEOREM 5.1. *Let $\epsilon \in (0, \frac{1}{2})$ be arbitrarily set in Algorithm 2. Let $\eta \in (0, \frac{1}{2})$ be such that $(\frac{1}{2} + \eta)(1 - \epsilon) > \frac{1}{2}$. Then there exists a constant $\lambda_0 < \infty$ depending on $f_{in}(\cdot)$, $f_{out}(\cdot)$, d , ϵ and η such that for all $\lambda > \lambda_0$, Algorithm 3 will solve Community Detection.*

REMARK 5.1. *An upper bound to the constant λ_0 is provided in the sequel in Proposition 5.6.*

To prove the main result, we will need an additional classification of the cells of B_n as either T-

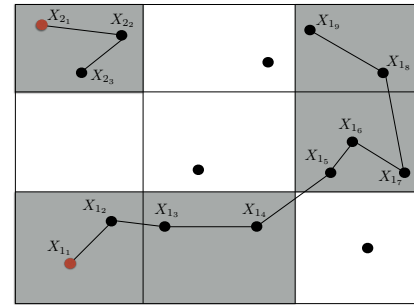


Figure 2: An illustration of algorithm 3. In this example, we do not draw the graph G , but only show the locations of the nodes. The shaded cells corresponds to A-Good cells and in this example there are two A-Good connected components. In each component, we outline an arbitrary sequence of points $X_{11} \cdots X_{19}$ and X_{21}, X_{22}, X_{23} that will be used in line 4 of our main Algorithm 3. The lines represent how we recursively set the community label estimates of the nodes as in line 8 of Algorithm 3.

Good or T-Bad. The nomenclature stands for *Truth-Good*.

DEFINITION 5.1. *A cell Q_z is **T-Good** if -*

1. $|\phi \cap Q_z| \geq \lambda \left(\frac{R}{4}\right)^d \frac{1}{d}(1 - \epsilon)$; and
2. For all $i, j \in \mathbb{N}$ such that $X_i, X_j \in Q_{L_1(z)} \cap \phi$, *Pairwise-Classify* (i, j, ϕ, G) returns $\mathbf{1}_{Z_i=Z_j} - \mathbf{1}_{Z_i \neq Z_j}$, i.e. the ground truth.

*If a cell is not T-Good, we call it **T-Bad**.*

A cell is T-Good, if for any pair nodes which either lie in the cell under consideration or the neighboring cells, the output of the pairwise estimation matches the ground-truth. Of-course since the ground truth is unknown, one cannot test whether a cell is T-Good or not. We introduce the notion of a T-Good cell to aid in the analysis.

The proof of Theorem 5.1 can be split into three parts. The first part is composed of combinatorial arguments leveraging the definitions of A-Good and T-Good cells. These combinatorial lemmas will conclude that it suffices to ensure that there exists a ‘giant’ T-Good connected component in the data (ϕ_n, G_n) . The next is a local analysis wherein we conclude that the probability a cell is T-Good can be made arbitrarily large by choosing the constant λ sufficiently high. The final step is to couple the

process of T-Good cells to that of dependent site percolation on $\mathbb{Z}^d$ to conclude that if a single cell is T-Good with sufficiently high probability, then there exists a giant T-Good component. Owing to space constraints, we only give the sketch here and refer to the full version [38] for details.

5.1 Combinatorial Analysis The main result in this sub-section we want to establish is the following statement. If we establish this, then the performance of our algorithm will follow from a study of the properties of the random graph G .

PROPOSITION 5.1. *If there exists a connected component of T-Good cells in the data (ϕ_n, G_n) which contains a fraction of nodes of G_n strictly larger than a half with probability $1 - o_n(1)$, then the output returned by Algorithm 3 solves the Community Detection problem as given in Definition 2.2.*

The proof is based on the following elementary combinatorial proposition, which is the basis of Line 5 in the GBG in Algorithm 3. For every $z \in \mathbb{Z}^d$, denote by $\mathcal{A}(z)$ the maximal $\mathbb{Z}^d$ connected set containing z such that for all $u \in \mathcal{A}(z)$, cell u is A-Good.

PROPOSITION 5.2. *For every $z \in \mathbb{Z}^d$ such that cell z is A-Good, there exists a unique partition of $\phi_{\mathcal{A}(z)} := \phi_{\mathcal{A}(z)}^{(+)} \coprod \phi_{\mathcal{A}(z)}^{(-)}$ such that for all $z, z' \in \mathcal{A}(z)$ with $\|z - z'\|_\infty \leq 1$ and all $X_i \in \phi \cap Q_z$ and $X_j \neq X_i \in \phi \cap Q_{z'}$, we have*

- If $X_i \in \phi_z^{(+)}$ and $X_j \in \phi_{z'}^{(-)}$ or, if $X_i \in \phi_z^{(-)}$ and $X_j \in \phi_{z'}^{(+)}$, then *Pairwise-Classify*(i, j, G) will return -1 .
- If $X_i, X_j \in \phi_z^{(+)}$ or if $X_i, X_j \in \phi_z^{(-)}$, then *Pairwise-Classify*(i, j, G) returns $+1$.

Moreover, the partition produced in Line 8 of our Algorithm 3 coincides with this partition.

This Proposition shows that all nodes inside A-Good connected components can be partitioned into two sets uniquely, such that the T-Good sub-component inside the A-Good component will be partitioned according to the underlying ground truth. Moreover, by following any arbitrary enumeration of the nodes of G as done in Line 5 of Algorithm 3, we can now build this unique partition of nodes of the A-Good component. This is what allows our algorithm to be fast. Using these two facts, we can conclude the proof of Proposition 5.1.

Proof. Proof of Proposition 5.1

Proposition 5.2 justifies Line 5 of Algorithm 3. First note that since every A-Good cell is non-empty of nodes of G , the arbitrary sequence in Line 5 of Algorithm 5.2 will enumerate all the nodes in each connected component. In other words, the only estimates that will be set in Line 13 of Algorithm 3 are those nodes that fall in the A-Bad cells. Moreover, the partition of the A-Good connected components in Line 8 will coincide with the partition referred to in Proposition 5.2. From the definition of T-Good components, the unique partition referred to in Proposition 5.2 will be such that the T-Good component will be partitioned according to the ground truth. Hence, if there exists a T-Good connected component that has a fraction $\alpha > \frac{1}{2}$ of the nodes of G_n , Algorithm 3 will partition this set of nodes in accordance to the ground truth. Thus, the achieved overlap will be at-least $2\alpha - 1 > 0$. This follows since the mis-classification of all nodes apart from this ‘giant’ connected T-Good component cannot diminish the overlap below $2\alpha - 1$ which is still positive.

5.2 Local Analysis The main goal of this subsection is to show that the probability a cell is T-Good can be made arbitrarily high by taking λ sufficiently high. In order to present the arguments, we will need the definition of a generalized Palm distribution. For any $k \in \mathbb{N}$ and $x_1, \dots, x_k \in \mathbb{R}^d$, we denote by $\mathbb{P}_\phi^{x_1, \dots, x_k}$ to be the Palm distribution of ϕ at $x_1, \dots, x_k$. This measure is the one induced by first sampling ϕ and G and then placing additional points at $x_1, \dots, x_k$ and equipping them with independent community labels and edges. More precisely, we give these nodes i.i.d. uniform community labels $Z_{-1}, \dots, Z_{-k} \in \{-1, 1\}^k$. Conditionally on all the labels and ϕ , we draw an edge between any $i, j \in \{-k, -(k-1), \dots\}$ such that at-least one of i or j belong to $\{-k, \dots, -1\}$ as before, i.e. with probability $f_{in}(\|X_i - X_j\|)$ if the two nodes have the same community labels or with $f_{out}(\|X_i - X_j\|)$ if the two nodes have opposite community labels independently of other edges.

PROPOSITION 5.3. *For any two $x \neq y \in \mathbb{R}^d$ such that $\|x - y\|_2 < 2R$, then conditionally on the labels of the points at x and y denoted as Z_x and Z_y respectively, we have for all $k \in \mathbb{N}$*

- If $Z_x = Z_y$, then $\mathbb{P}^{x,y}[E_G^{(R)}(x, y) = k] = \frac{e^{-\lambda M_{in}(x,y)} (\lambda M_{in}(x,y))^k}{k!}$, i.e. is distributed as a Poisson random variable with mean $\lambda M_{in}(x, y)$.
- If $Z_x \neq Z_y$, then $\mathbb{P}^{x,y}[E_G^{(R)}(x, y) = k] =$

$\frac{e^{-\lambda M_{out}(x,y)} (\lambda M_{out}(x,y))^k}{k!}$, i.e. is distributed as a Poisson random variable with mean $\lambda M_{out}(x,y)$.

This follows directly from Slivnyak's theorem and the independent thinning property of PPPs. We can then leverage this to prove the following result.

PROPOSITION 5.4. *For all connection functions $f_{in}(\cdot)$ and $f_{out}(\cdot)$ satisfying the hypothesis of Theorem 3.2, there exists a constant $c > 0$ such that for all $x \neq y \in \mathbb{R}^d$ satisfying $\|x - y\|_2 \leq (3/4)R$, we have*

$$\mathbb{P}^{x,y}[(x,y) \text{ is misclassified by Algorithm 1}] \leq e^{-c\lambda},$$

where the constant c satisfies

$$c \geq \inf_{x,y \in \mathbb{R}^d: \|x-y\|_2 \leq 3R/4} (\mathbf{1}_{M_{out}(x,y)>0} M_{out}(x,y) + \mathbf{1}_{M_{out}(x,y)=0} M_{in}(x,y)) h \left(\frac{M_{in}(x,y) - M_{out}(x,y)}{2M_{in}(x,y)} \right),$$

where $h(t) := (1+t) \log(1+t) - t$, for all $t \in \mathbb{R}_+$. In particular, $c > 0$ is strictly positive.

This result follows from an elementary application of the Chernoff's bound for Poisson random variables. These two results can be put together to derive the following useful proposition.

LEMMA 5.1. *For all $z \in \mathbb{Z}^d$,*

$$\mathbb{P}[\text{Cell } z \text{ is T-Good in graph } G] \geq 1 - e^{-\lambda(R/4)^d \frac{1}{d} h(\epsilon)} - \lambda^2 (3R/4)^d \frac{1}{d} e^{-c\lambda},$$

where the constant c and function $h(\cdot)$ are defined in Proposition 5.4.

This follows from an application of the union bound. From properties of PPP, the probability that a cell has fewer than $\lambda(R/4d^{1/d})^d(1-\epsilon)$ is exponentially (in λ) small. To bound the probability that there are no pairs of points which fail **Pairwise-Classify**, it suffices from the first moment method to compute their expectation. This in turn can be computed by an application of the second order moment measure expansion of the PPP [9]. Thus, we immediately have the following corollary which is what we will use in the sequel.

COROLLARY 5.1. *For every $p \in (0,1)$, and every $f_{in}(\cdot)$ and $f_{out}(\cdot)$ satisfying the hypothesis of Theorem 3.2, there exists a λ' such that for all $\lambda > \lambda'$, and all $z \in \mathbb{Z}^d$, $\mathbb{P}[\text{Cell } z \text{ is T-Good}] \geq p$.*

5.3 Global Analysis In this section, we present the central tool required to analyze about the 'giant' connected T-Good component in the graph G_n . To do so, we exploit a coupling between the T-Good cells in the graph G and a certain dependent site percolation process on $\mathbb{Z}^d$. Denote by $(Y_z)_{z \in \mathbb{Z}^d}$ to be the random 0 – 1 field on $\mathbb{Z}^d$ where $Y_z := \mathbf{1}_{\text{Cell } z \text{ is T-Good in } G}$. From the construction of the field, notice that the random field $(Y_z)_{z \in \mathbb{Z}^d}$ is only mildly dependent. Indeed, given any two $z, z' \in \mathbb{Z}^d$, such that $\|z - z'\|_1 \geq 12d^{1/d}$, we have that Y_z and $Y_{z'}$ are independent random variables. This follows from the fact that we only look upto Euclidean distance of at-most $2R$ from any point inside a cell z to determine whether a cell is T-Good or T-Bad. Since, in an independently marked PPP, events corresponding to disjoint sets of $\mathbb{R}^d$ are independent, the claim follows.

We will now set some notation that will be useful in studying the process $(Y_z)_{z \in \mathbb{Z}^d}$. For any $z \in \mathbb{Z}^d$, cell z is **open** in $\mathbb{Z}^d$ if $Y_z = 1$. Similarly, any edge connecting z and z' is said to be open if both its end points are open. For any $z \in \mathbb{Z}^d$, we denote by $\mathcal{C}(z)$ to be the maximal connected random subset of $\mathbb{Z}^d$ containing z such that all $z' \in \mathcal{C}(z)$ satisfies $Y_{z'} = 1$. The main proposition we want to establish in this section is the following.

PROPOSITION 5.5. *For every $\eta \in (0, \frac{1}{2})$, there exists $\lambda_0(\eta, \epsilon) < \infty$ (where ϵ is set in Algorithm 2) chosen sufficiently high (as a function of $f_{in}(r), f_{out}(r), r \in [0, R]$ and d), such that for all $\lambda > \lambda_0(\eta, \epsilon)$ and all $j \in \mathbb{Z}^d$*

$$(5.2) \quad \liminf_{n \rightarrow \infty} \frac{1}{(2n)^d} \sum_{i \in \mathbb{Z}^d: \|i-j\|_\infty \leq n} \mathbf{1}_{i \in \mathcal{C}(j)} \geq \frac{1}{2} + \eta,$$

$\mathbb{P}$ almost-surely on the event that $\{|\mathcal{C}(j)| = \infty\}$. Moreover, for $\lambda > \lambda_0(\eta, \epsilon)$, and all $j \in \mathbb{Z}^d$, $\mathbb{P}[|\mathcal{C}(j)| = \infty] \geq \frac{1}{2} + \eta$ and $\mathbb{P}[\exists j \in \mathbb{Z}^d: |\mathcal{C}(j)| = \infty] = 1$.

The key insight out of the proposition we want is to ensure that by taking λ sufficiently high, there exists an infinite open component in the process $(Y_z)_{z \in \mathbb{Z}^d}$, i.e. there exists $z \in \mathbb{Z}^d$ such that $|\mathcal{C}(z)| = \infty$. Moreover, we want to show that this infinite component contains more than half of the sites of $\mathbb{Z}^d$. The reason this does not immediately follow from Corollary 5.1 is that we have not yet established that the infinite open component in $(Y_z)_{z \in \mathbb{Z}^d}$ if it exists is unique. However [21] provides a clean 'black-box' methodology to establish this and our proposition can be viewed as a direct corollary of Theorem 1 in [21]. We will first dominate the process $(Y_z)_{z \in \mathbb{Z}^d}$ by an

independent percolation process which is known to have a unique infinite component and then leverage this domination to conclude the proposition.

The proof of Theorem 5.1 now follows from ergodicity of site percolation process. Namely, if the percolation probability of site percolation on $\mathbb{Z}^d$ is $q \in (0, 1)$, then with probability $1 - o_n(1)$, the largest connected component of the percolation process restricted to any box of side $n^{1/d}$ contains at least $(q - \delta)n$ number sites in that box for any small $\delta > 0$.

5.4 An Upper Bound to the Constant $\lambda(\epsilon, \eta)$

PROPOSITION 5.6. *If λ satisfies*

$$\frac{1}{3} \sum_{n \geq M} n \left(3 \left(e^{-\lambda(R/4d^{1/d})^d h(\epsilon)} + \lambda^2 \left(\frac{3R}{4} \right)^d \frac{1}{d} e^{-c\lambda} \right)^{1/M} \right)^n \leq \frac{1}{2} - \eta,$$

where the constant c is given in Proposition 5.4, $M := \lceil 12d^{1/d} \rceil$, ϵ is set in Algorithm 2 and $h(x) := (1+x) \log(1+x) - x$, then $\lambda \geq \lambda_0(\epsilon, \eta)$.

The proof follows from standard Pierl's argument [6].

6 Lower Bound for Community Detection

The goal of this section is to prove Theorem 3.1. The proof follows in two steps. First, we show in Proposition 6.1 that Community Detection is easier than a certain problem we call Information Flow from Infinity (defined in Definition 6.1). Subsequently, we provide an impossibility for Information flow From Infinity in Section 6.3. But before presenting the proof, we discuss the proof of Proposition 3.1 (which was stated in Section 3), which identifies a special case of connection functions $f_{in}(\cdot)$ and $f_{out}(\cdot)$ when the phase-transition is sharp. This will provide the intuition for our lower bound.

6.1 Proof of Proposition 3.1 Let $R_{in} > R_{out} \geq 0$ be arbitrary and consider the two functions to be $f_{in}(r) = \mathbf{1}_{r \leq R_{in}}$ and $f_{out}(r) = \mathbf{1}_{r \leq R_{out}}$. In words, two points of opposite communities are connected if and only if their distance is lesser than R_{out} and two points of the same community are connected if and only if their distance is smaller than R_{in} . In this example, it is clear that for any two points $X_i, X_j \in \phi$, no matter their community labels Z_i and Z_j , if $\|X_i - X_j\|_2 \leq R_{out}$, then i and j are always connected in G . Similarly, any two points X_i and X_j such that $\|X_i - X_j\|_2 > R_{in}$ are never connected by an edge

in G no matter their community labels Z_i and Z_j . Hence, the *informative* pairs of points in this example are those X_i, X_j such that $\|X_i - X_j\|_2 \in (R_{out}, R_{in}]$. Moreover, it is immediate that, if $\|X_i - X_j\|_2 \in (R_{out}, R_{in}]$ and $i \sim_G j$, then $Z_i = Z_j$. On the other hand if $\|X_i - X_j\|_2 \in (R_{out}, R_{in}]$ and $i \not\sim_G j$, then it must be the case that $Z_i \neq Z_j$. For any two points X_i and X_j such that $\|X_i - X_j\|_2 \in [0, R_{out}] \cup (R_{in}, \infty)$, the presence or absence of an edge is not informative as it is a certain event.

This example motivates the following algorithm for Community Detection. Partition the nodes of G_n into $\mathcal{D}_1, \dots, \mathcal{D}_k$ where each component $\mathcal{D}_i$ is a maximal set of nodes $\{X_{i_1}, \dots, X_{i_{l_i}}\}$ of G_n such that for all $j \in [1, l_i]$, we have $\|X_{i_{j-1}} - X_{i_j}\| \in (R_{out}, R_{in}]$. In words, we form another graph T_n from the points ϕ_n such that any two nodes i and j of G_n are connected in T_n if and only if $\|X_i - X_j\| \in (R_{out}, R_{in}]$. Then $\mathcal{D}_1, \dots, \mathcal{D}_k$ are the connected components of the graph T_n . The algorithm works by considering and labeling each connected component $\mathcal{D}_i$ independently of other components. For each cluster $i \in [1, k]$, estimate the node label of X_{i_1} to be $+1$. Then for every $j \in [2, l_i]$, recursively estimate the node label by the following procedure-

- If $i_{j-1} \sim_{G_n} i_j$ then set $Z_{i_j} = Z_{i_{j-1}}$.
- If $i_{j-1} \not\sim_{G_n} i_j$ then set $Z_{i_j} = -Z_{i_{j-1}}$.

This algorithm considers each of the connected component of T_n enumerated in an arbitrary manner and then labels the nodes in these components.

This algorithm partitions each component $\mathcal{D}_i$, $i \in [1, k]$ exactly in accordance to the ground truth. However, it could be that the estimated signs in each of the connected components $\mathcal{D}_i$ could be flipped from the underlying ground truth and hence the achieved overlap can still be small even though we partition each cluster $\mathcal{D}_i$ accurately. To argue that the overlap achieved is not too small, a sufficient condition is that there exists a unique giant (of size $cn - o(n)$ for some $c > 0$) component of T_n and all other connected components are $o(n)$. Then, we will have by the strong-law of large numbers that the overlap achieved will be c , i.e. the mislabeling in all small components will 'cancel' each other out and in particular cannot drive the overlap of c achieved in the giant component to 0.

From the definition of percolation, a unique giant component in T_n exists if and only if $\theta(H_{\lambda, f_{in}(\cdot) - f_{out}(\cdot), d}) > 0$ since $T_n \stackrel{(d)}{=} H_{\lambda, f_{in}(\cdot) - f_{out}(\cdot), d}$.

In the sequel, we will generalize the above example to come up with the general lower bound for Community Detection problem.

6.2 Information Flow from Infinity This problem refers to how well can one estimate the community label of a tagged node of a graph better than at random, given some extra ‘information at infinity’. We set some notation first to precisely define the problem. We will pose this question under the Palm Probability measure $\mathbb{P}^0$. Recall that the Palm measure is the distribution of the graph G ‘seen from a typical node’. Recall that under this measure, there is an additional node at origin, with an independent community label denoted by $Z_0 \in \{-1, +1\}$. For every $r \in \mathbb{R}_+$, denote by $\phi^{(r)}$ and $G^{(r)}$ the point-process and graph, in which every vertex $i \in \mathbb{N}$ (which is at location $X_i \in \mathbb{R}^d$) is equipped with the random variable $Z_i \mathbf{1}_{\|X_i\|_2 \geq r}$. Note that this is not a mark since it is not translation invariant, but is a random variable associated with vertex i . In words, we retain the community label marks on nodes of G at a Euclidean distance of r or more from the origin and delete (i.e. set to 0) the community label of those nodes which are located at distances less than r from the origin.

DEFINITION 6.1. *We say Information Flows from Infinity if for every $r \in \mathbb{R}_+$ there exists a random variable $\tau_r \in \{-1, +1\}$, measurable (deterministic function) with respect to the observed data $(\phi^{(r)}, G^{(r)})$ and a constant $\gamma > 0$ such that*

$$(6.3) \quad \liminf_{r \rightarrow \infty} \mathbb{P}^0[\tau_r = Z_0] \geq \frac{1}{2} + \gamma.$$

We say information ‘flows’ from infinity if we are able to non-trivially estimate the community label at origin, given ‘information at infinity’. Note that for each r , there exists algorithms (i.e. τ_r) such that $\mathbb{P}^0[\tau_r = Z_0] > \frac{1}{2}$. However, the non-trivial question is to understand if the limit as $r \rightarrow \infty$ is still strictly larger than a half. This definition is similar in spirit to those considered in Ising models to detect phase-transition for multiplicity of Gibbs states (as in [4] and [33]).

Before analyzing it, we establish monotonicity property the problem enjoys.

PROPOSITION 6.1. *For every $d \in \mathbb{N}$ and $f_{in}(\cdot), f_{out}(\cdot) : \mathbb{R}_+ \rightarrow [0, 1]$, the limit $\lim_{r \rightarrow \infty} \sup_{\tau_r \in \sigma((\bar{G}^{(r)}, \bar{\phi}^{(r)})} \mathbb{P}^0[\tau_r = Z_0]$ exists. Moreover, $\lambda \rightarrow \lim_{r \rightarrow \infty} \sup_{\tau_r \in \sigma((\bar{G}^{(r)}, \bar{\phi}^{(r)})} \mathbb{P}^0[\tau_r = Z_0]$ is non-decreasing.*

Note the supremum is over all possible estimators of the community label at origin. We skip the proof as it follows from straightforward arguments of independent thinning properties of the PPP. Refer to [38] for details. In what follows, we will need the following classical result on the ergodic property of marks of a stationary point process.

PROPOSITION 6.2. ([11]) *Let $\phi := \{X_1, X_2, \dots\}$ be a homogeneous PPP with its atoms enumerated in an arbitrary measurable way. Let each atom $i \in \mathbb{N}$ be assigned a translation invariant mark random variable $J_i \in \Xi$ taking values in an arbitrary Borel measurable space $(\Xi, \mathfrak{g})$. Let $B_n := \left[-\frac{n^{1/d}}{2}, \frac{n^{1/d}}{2}\right]^d$ be the box of volume n and let $X_{j(n)} \in \phi$ be chosen uniformly at random among the atoms of ϕ that lie in B_n , if any. Then for all $A \in \mathfrak{g}$, the limit $\lim_{n \rightarrow \infty} \mathbb{P}[J_{j(n)} \in A]$ exists and satisfies $\lim_{n \rightarrow \infty} \mathbb{P}[J_{j(n)} \in A] = \mathbb{P}^0[J_0 \in A]$, where J_0 is the mark of the atom of ϕ at origin under $\mathbb{P}^0$.*

The following proposition establishes that Community Detection is harder than Information Flow from Infinity.

LEMMA 6.1. *If there exists a Community Detection algorithm (polynomial or exponential time) that achieved an overlap of $\gamma > 0$, then $\lim_{r \rightarrow \infty} \sup_{\tau_r \in \sigma((\bar{G}^{(r)}, \bar{\phi}^{(r)})} \mathbb{P}^0[\tau_r = Z_0] \geq \frac{1+\gamma}{2}$.*

We skip the details and only provide an intuition here. It is easy to see that one can determine the membership of two nodes chosen uniformly at random better than chance, if one can cluster the entire graph in a way that it has positive overlap with the ground truth. Thus, Community detection is harder than estimating the community membership of two nodes chosen at random. Now, with high probability, the two chosen nodes will be ‘far’: in particular their distance will exceed any constant r with high probability. Thus, we can make this pairwise problem easier by picking one of the nodes, and revealing the true community membership of all other nodes at a distance r away from this. From the Theorem 6.2, this is equal to the Palm probability of identifying the typical node’s membership given the true community labels at distance r away from it. Now, r can be any large constant, this implies that if one cannot solve Information Flow from Infinity problem, then it is impossible to determine better than chance, the community membership of two nodes chosen uniformly at random.

The following is the main technical result on the Information Flow from Infinity problem.

THEOREM 6.1. *For every $\lambda, f_{in}(\cdot), f_{out}(\cdot)$ and d , the following limit exists and satisfies*

$$(6.4) \quad \lim_{r \rightarrow \infty} \sup_{\tau_r \in \sigma(G^{(r)}, \phi^{(r)})} \mathbb{P}^0[\tau_r = Z_0] \leq \frac{1}{2} (1 + \theta(H_{\lambda, f_{in}(\cdot) - f_{out}(\cdot), d})).$$

Recall that $\theta(H_{\lambda, f_{in}(\cdot) - f_{out}(\cdot), d})$ is the percolation probability of the classical random connection model where any two nodes of ϕ located at $x, y \in \mathbb{R}^d$ are connected by an edge with probability $f_{in}(\|x - y\|) - f_{out}(\|x - y\|)$. The supremum is over all valid estimators of the community label at origin and hence if $\theta(H_{\lambda, f_{in}(\cdot) - f_{out}(\cdot), d}) = 0$, then there is no estimator that will solve the Information Flow from Infinity problem. In view of Lemma 6.1, we also get that if $\theta(H_{\lambda, f_{in}(\cdot) - f_{out}(\cdot), d}) = 0$, then there is no algorithm (polynomial or exponential time) to solve Community Detection. Thus, if we prove Theorem 6.1, then we will conclude the proof of Theorem 3.1.

Before presenting the proof of Theorem 6.1, we illustrate a few example setting where the bound in Equation 6.4 is tight and loose respectively. In view of Lemma 6.1 and Proposition 3.1, the following corollary where Equation (6.4) is tight holds.

COROLLARY 6.1. *For all $\lambda > 0, R_1 \geq R_2$, if $f_{in}(r) = \mathbf{1}_{r \leq R_1}$ and $f_{out}(r) = \mathbf{1}_{r \leq R_2}$*

$$\lim_{r \rightarrow \infty} \sup_{\tau_r \in \sigma(G^{(r)}, \phi^{(r)})} \mathbb{P}^0[\tau_r = Z_0] = \frac{1}{2} (1 + \theta(H_{\lambda, f_{in}(\cdot) - f_{out}(\cdot), d})).$$

In other words, we see that the inequality in Theorem 6.1 is achieved in certain examples. However, Theorem 6.1 is not an accurate characterization of the Information Flow from Infinity problem as evidenced in the following example.

PROPOSITION 6.3. *For all $d \geq 2$, if $f_{in}(r) = \min\left(1, \frac{1}{\sqrt{r}} + \frac{1}{r^{d-1/4}}\right)$ and $f_{out}(r) = \min\left(1, \frac{1}{\sqrt{r}}\right)$, the inequality in Equation (6.4) is strict for all values of $\lambda > 0$.*

The proof of this follows from absolute continuity of infinite measures [41]. Refer to [38] for the details.

6.3 The Information Graph and Proof of Theorem 6.1 The main goal in this section is to construct an ‘information graph’ denoted by I and

conclude that if this graph does not percolate, then one cannot solve the Information Flow from Infinity problem. The graphs I and G are coupled and built on the same probability space using the same set of random elements. The vertex set of I is $\mathbb{N}$. The nodes are only equipped with the location labels $\{X_i\}_{i \in \mathbb{N}}$ and not the community labels. The edges are based on the random elements $\{\{U_{ij}\}_{j>i}\}_{i \in \mathbb{N}}$, which are the same used to construct G . We connect any $i < j \in \mathbb{N}$ by an edge in graph I if $U_{ij} \in [f_{out}(\|X_i - X_j\|), f_{in}(\|X_i - X_j\|)]$. We denote by $i \sim_I j$ the event that points i and j are connected by an edge in I . Recall that the graph G was built by connecting any two points $i < j \in \mathbb{N}$ if $U_{ij} \leq \mathbf{1}_{Z_i=Z_j} f_{in}(\|X_i - X_j\|) + \mathbf{1}_{Z_i \neq Z_j} f_{out}(\|X_i - X_j\|)$. For each $i \in \mathbb{N}$, we denote by $V_I(i) \subseteq \mathbb{N}$ the random subset of the nodes contained in the connected component of node i in graph I . Observe that the information graph $I \stackrel{(d)}{=} H_{\lambda, f_{in}(\cdot) - f_{out}(\cdot), d}$, i.e. the I graph we constructed is equal in distribution to the graph of the Poisson Random Connection model with vertex set forming a PPP of intensity λ and connecting any two vertices at distance r away with probability $f_{in}(r) - f_{out}(r)$ independently of everything else.

LEMMA 6.2. *From the way we have coupled the construction of G and I , we have*

- If $i \sim_I j$ and $i \sim_G j$, then $Z_i = Z_j$.
- If $i \sim_I j$ and $i \not\sim_G j$, then $Z_i \neq Z_j$.

Proof. This follows from the following construction of G and I as follows.

- $i \sim_G j$ if and only if $U_{ij} \leq f_{in}(\|X_i - X_j\|) \mathbf{1}_{Z_i=Z_j} + f_{out}(\|X_i - X_j\|) \mathbf{1}_{Z_i \neq Z_j}$.
- $i \sim_I j$ if and only if $U_{ij} \in [f_{out}(\|X_i - X_j\|), f_{in}(\|X_i - X_j\|)]$.
- $\forall r \geq 0, f_{in}(r) \geq f_{out}(r)$.

The lemma follows since the $\{U_{ij}\}_{0 \leq i < j}$ are the same with which we build both the graphs G and I .

We can iterate the above lemma from edges to connected components of I .

LEMMA 6.3. *For all $j \in \mathbb{N}$, conditional on G, ϕ, I , there are exactly two possible sequences $(Z_k)_{k \in V_I(j)}$ which are complements of each other that are consistent in the sense of Lemma 6.2 with the observed data G, ϕ and I .*

The proof is elementary and we skip it for brevity. The following lemma, now essentially says the community labels on disconnected components of I are independent.

LEMMA 6.4. *For all $\lambda > 0$, on the event $\{|V_I(0)| < \infty\}$,*

$$\mathbb{P}^0 \left[Z_0 = +1 \middle| G, \{\{U_{kl}\}_{l>k}\}_{k \in \mathbb{N} \cup \{0\}}, \phi, \{Z_i\}_{i \in V(I_{(0)})^c} \right] = \frac{1}{2} \text{ a.s.}$$

The proof follows from an application of Lemma 6.3 and Bayes' rule. The key reason why we need the condition $\{|V_I(0)| < \infty\}$ is to be able to apply the Bayes' rule without encountering a 0 by 0 situation. With this construction, we can now conclude the proof of Theorem 6.1 by conditioning on the fact that $V_I(0)$ is contained within a ball of radius r . On this event, the above lemma gives that we cannot estimate the community label at origin better than chance. On the complement of this condition, we can upper bound the probability of estimating the community label at origin by 1. But since $\theta(H_{\lambda, f_{in}(\cdot) - f_{out}(\cdot), d}) = 0$, there exists an almost-surely finite r such that $V_I(0)$ is contained within a ball of radius r , and hence the Information Flow from Infinity cannot be solved. The details can be found in [38].

7 Identifiability of the Partition and Proof of Theorem 3.3

The key technical tool is the ergodicity of the PPP which is summarized in the following lemma. We need to set some notation that are needed to state the lemma. Denote by $\mathbb{M}_{\Xi}(\mathbb{R}^d)$ the set of all 'marked' point processes on $\mathbb{R}^d$ where each point is assigned a 'mark' from the measure space Ξ , with its associated sigma-algebra. The set $\mathbb{M}_{\Xi}(\mathbb{R}^d)$ is a Polish space, has a natural topology and hence an associated sigma-algebra (see [11]). Denote by $\theta : \mathbb{R}^d \times \mathbb{M}_{\Xi}(\mathbb{R}^d) \rightarrow \mathbb{M}_{\Xi}(\mathbb{R}^d)$ the 'shift' operator which is a measurable function where $\theta(x, \psi)$ retains the same marks but translates all points of ψ by a vector x .

LEMMA 7.1. *Let $C_n \subset \mathbb{R}^d$ be a sequence of L_p , $p \in [1, \infty]$ balls centered at the origin with radius going to infinity as $n \rightarrow \infty$. Let $f : \mathbb{M}_{\Xi}(\mathbb{R}^d) \rightarrow \mathbb{R}_+$ be a measurable function such that $\mathbb{E}_{\phi}^0[f] < \infty$. Then,*

the following limit exists :

$$(7.5) \quad \lim_{n \rightarrow \infty} \frac{\sum_{i \in \mathbb{N}} \mathbf{1}_{X_i \in C_n} f \circ \theta(X_i, G)}{\sum_{i \in \mathbb{N}} \mathbf{1}_{X_i \in C_n}} = \mathbb{E}^0[f] \text{ } \mathbb{P} \text{ a.s. .}$$

We now prove Theorem 3.3.

Proof. First if $g(\cdot) \neq \frac{f_{in}(\cdot) + f_{out}(\cdot)}{2}$, then G and $H_{\lambda, g(\cdot), d}$ are mutually singular and this can be seen through the following elementary argument. Fix some $L < \infty$ such that $\int_{x \in \mathbb{R}^d: \|x\| \leq L} g(\|x\|) dx \neq \int_{x \in \mathbb{R}^d: \|x\| \leq L} ((f_{in}(x) + f_{out}(\|x\|))/2) dx$ are both finite. Such a L exists since $g(\cdot) \neq \frac{f_{in}(\cdot) + f_{out}(\cdot)}{2}$. Now, we apply the ergodic theorem, where every node $i \in \mathbb{N}$ of ϕ is equipped with a mark $\Xi_i \in \mathbb{N}$, which denotes the number of graph neighbors of node i at a distance of at-most L from X_i , i.e. $\Xi_i = |\{j \in \mathbb{N} \setminus \{i\} : i \sim_G j, \|X_i - X_j\| \leq L\}|$. Thus, the Ergodic theorem implies that the measure induced by G will be concentrated on the set

$$\left\{ g \in \mathbb{M}_G(\mathbb{R}^d) : \lim_{n \rightarrow \infty} \frac{\sum_{i, j \in \mathbb{N}} \mathbf{1}_{\|X_i\| \leq n, \|X_i - X_j\| \leq L} \mathbf{1}_{i \sim_{g,j}}}{\sum_{i, j \in \mathbb{N}} \mathbf{1}_{\|X_i\| \leq n, \|X_i - X_j\| \leq L}} = \int_{x \in \mathbb{R}^d: \|x\| \leq L} (f_{in}(x) + f_{out}(\|x\|))/2 dx \right\},$$

while the measure induced by $H_{\lambda, g(\cdot), d}$ will be concentrated on the set

$$\left\{ g \in \mathbb{M}_G(\mathbb{R}^d) : \lim_{n \rightarrow \infty} \frac{\sum_{i, j \in \mathbb{N}} \mathbf{1}_{\|X_i\| \leq n, \|X_i - X_j\| \leq L} \mathbf{1}_{i \sim_{g,j}}}{\sum_{i, j \in \mathbb{N}} \mathbf{1}_{\|X_i\| \leq n, \|X_i - X_j\| \leq L}} = \int_{x \in \mathbb{R}^d: \|x\| \leq L} g(\|x\|) dx \right\}.$$

Thus, the only case to consider is the one where $g(\cdot) = (f_{in}(\cdot) + f_{out}(\cdot))/2$. From linearity of expectation, the average degree of any node $i \in \mathbb{N}$ in both graphs G and $H_{(\lambda, \frac{f_{in}(\cdot) + f_{out}(\cdot)}{2}, d)}$ is the same and equal to $(\lambda/2) \int_{x \in \mathbb{R}^d} (f_{in}(\|x\|) + f_{out}(\|x\|)) dx$ and thus empirical average of the degree does not help. However, we see that the triangle profiles differ in the two models which we leverage to prove the Theorem. For ease of notation, we denote by $H := H_{\lambda, (f_{in}(\cdot) + f_{out}(\cdot))/2, d}$. Let $h(x, y) = \mathbf{1}_{\|x\| \leq R} \mathbf{1}_{\|y\| \leq R} \mathbf{1}_{f_{in}(\|x\|) > f_{out}(\|y\|)} \mathbf{1}_{f_{in}(\|y\|) > f_{out}(\|x\|)} \mathbf{1}_{f_{in}(\|x-y\|) > f_{out}(\|x-y\|)}$, for some $R > 0$ such that $0 < \int_{y \in \mathbb{R}^d} h(0, y) dy < \infty$. Define

$$\Delta = \mathbb{E}^0 \left[\sum_{x \neq y \neq 0} h(x, y) \mathbf{1}_{((0, x) \in E, (0, y) \in E, (x, y) \in E)} \right].$$

Denote by Δ_G and Δ_H the value of the above expression if the underlying graphs were G and H respectively. From the moment measure expansion of PPPs [9], we get that

$$\Delta_G = \int_{x \in \mathbb{R}^d} \int_{y \in \mathbb{R}^d} h(x, y) \left(f_{in}(\|x - y\|) \left(\frac{f_{in}(\|x\|)f_{in}(\|y\|) + f_{out}(\|x\|)f_{out}(\|y\|)}{4} \right) + \right. \\ \left. f_{out}(\|x - y\|) \left(\frac{f_{in}(\|x\|)f_{out}(\|y\|) + f_{out}(\|x\|)f_{in}(\|y\|)}{4} \right) \right) \lambda^2 dx dy,$$

and

$$\Delta_H = \int_{x \in \mathbb{R}^d} \int_{y \in \mathbb{R}^d} h(x, y) \left(\frac{f_{in}(\|x\|) + f_{out}(\|x\|)}{2} \right) \left(\frac{f_{in}(\|y\|) + f_{out}(\|y\|)}{2} \right) \\ \left(\frac{f_{in}(\|x - y\|) + f_{out}(\|x - y\|)}{2} \right) \lambda^2 dx dy.$$

Now observe that

$$\Delta_G - \Delta_H = \int_{x \in \mathbb{R}^d} \int_{y \in \mathbb{R}^d} h(x, y) \\ \frac{1}{2} (f_{in}(\|x\|) - f_{out}(\|x\|)) (f_{in}(\|y\|) - f_{out}(\|y\|)) \\ (f_{in}(\|x - y\|) - f_{out}(\|x - y\|)) \lambda^2 dx dy.$$

From the fact that $f_{in}(r) \geq f_{out}(r)$ and $f_{in}(r)$ is not equal to $f_{out}(r)$ Lebesgue almost everywhere, there exists a $R < \infty$ to define $h(\cdot, \cdot)$ such that $0 \leq \Delta_H < \Delta_G < \infty$. Choose such a R and consider the following estimator:

Algorithm 4 Detect-Partitions

Given the data, i.e. locations of nodes and the graph, pick a L large enough and compute

$$\Delta^{(L)} := \frac{\sum_{i \in \mathbb{N}} \mathbf{1}(|X_i| \leq L) \tilde{h}(X_i)}{\sum_{i \in \mathbb{N}} \mathbf{1}(|X_i| \leq L)},$$

where $\tilde{h}(X_i) = \sum_{j, k \in \mathbb{N}, j \neq k \neq i} h(X_i - X_j, X_i - X_k) \mathbf{1}(i \sim_G j, i \sim_G k, j \sim_G k)$.

From ergodicity, we know that $\lim_{L \rightarrow \infty} \Delta^{(L)} = \Delta_G$, $\mathbb{P}$ almost-surely if the data is the block model graph or $\lim_{L \rightarrow \infty} \Delta^{(L)} = \Delta_H$ $\mathbb{P}$ almost surely if the graph is drawn according to the null model. We can

apply the ergodic theorem since a spatial random graph is a marked point process (as described in Section 2.1). Thus, the measures induced by (ϕ, G) and $(\phi, H_{\lambda, \frac{f_{in}(\cdot) + f_{out}(\cdot)}{2}, d})$ are mutually singular. Moreover, the above algorithm when tested on the finite data (G_n, ϕ_n) runs in time proportional to λn with the multiplicative constants depending on the connection functions $f_{in}(\cdot)$ and $f_{out}(\cdot)$ with success probability $1 - o_n(1)$ with the $o_n(1)$ term depending on the function $h(\cdot, \cdot)$ chosen.

We investigated the singularity of measures in order to understand the question of distinguishability of the planted partition model. This is a hypothesis testing problem of whether the data (graph and spatial locations) is drawn from the distribution of G or from the distribution of $H_{\lambda, g(\cdot), d}$ with a probability of success exceeding a half given a uniform prior over the models. This problem is in some sense easier than Community Detection, since this asks for asserting whether a partition exists or not, which intuitively should be simpler than finding the partition. Indeed, we show this in our model by proving that the distinguishability problem is trivially solvable while community detection undergoes a phase transition and is solvable only under certain regimes. In the general sparse SBM however, the equivalence between distinguishability and community detection is only conjectured and not yet proven ([12], [3]).

8 Conclusions and Open Problems

In this paper, we introduced the problem of community detection in a spatial random graph where there are two equal sized communities. Our main technical contributions are in identifying the problem of Information Flow from Infinity and connecting that with the Community Detection problem and giving a simple lower bound criterion. For developing the algorithm, we noticed that a spatial graph is sparse due to the fact that all interactions are dense, but localized which is starkly different from the reason why an Erdős-Rényi graph is sparse. We leveraged this difference to propose an algorithm for community detection by borrowing further ideas from dependent site percolation processes. However, this is just a first step and there are a plenty of open questions just concerning the model we introduced.

1) *Are Community Detection and Information Flow from Infinity equivalent?* - We proved that Community Detection was harder than Information Flow from Infinity. However, our algorithm and its

analysis showed that it can solve Community Detection whenever it can solve Information Flow from Infinity. Thus a natural question is whether these two problems undergo a phase-transition at the same point? Moreover is there a relation between the optimal overlap achievable in Community Detection and the optimal success probability of estimating the community label of the origin in the Information Flow from Infinity problem?

2) *Is the optimal overlap in Community Detection Monotone?* - We only established in Proposition 2.1 that solvability is monotone and not the optimal overlap achievable.

3) *More than 2 Communities* - In the symmetric case where the connection function is $f_{in}(\cdot)$ within communities and $f_{out}(\cdot)$ across communities, a simple adaptation of our algorithm can give a sufficient condition, although will be sub-optimal. Our lower bound technique can also be applied in the setting of many communities (see Theorem 2 in [2] for example) thereby establishing the *existence* of a non-trivial phase transition for any number of communities in the symmetric setting. But the open question is to identify examples similar to Proposition 3.1 where the phase transition can be tight. A quest for such examples can possibly lead to better understanding of even the 2 community case. Moreover, unified algorithmic techniques capable of handling non-symmetric case also is of interest in many practical settings.

4) *Characterization of the Phase-Transition* - An obvious but harder question is whether one can characterize if not compute the exact phase-transition for either Community Detection or Information Flow from Infinity. We show that our lower bound is capturing the phase-transition only in very specific cases and may not be tight in general due to corner cases similar to Proposition 6.3. We also have no reason to believe that our algorithm is optimal in any sense. Thus, a structural characterization of the phase-transition is still far from being understood.

5) *Computational Phase-Transition* - Another deep issue is computational complexity. Is there a regime where Community Detection is solvable, but no polynomial (in n) time and space algorithms that operate on (G_n, ϕ_n) are known to exist?

6) *Estimating the Model Parameters* - How does

one efficiently estimate the connection functions $f_{in}(\cdot)$ and $f_{out}(\cdot)$ from the data of just the graph G and the spatial locations ϕ .

Acknowledgements

This work was supported by a grant of the Simons Foundations (#197982 to The University of Texas at Austin). The authors thank numerous discussions with Emmanuel Abbe for suggesting the problem and pointing out the literature. The authors also thank Gustavo de Veciana, Sanjay Shakkottai, Joe Neeman and Alex Dimakis for providing comments on an earlier version of this work. We also thank the anonymous reviewers for constructive comments.

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